Faculty of Computers and Artificial Intelligence

Cairo University

Second Level - Second Semester

Software Engineering (CS251)

A1 Task (1,2,3)

Team Members

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Summary of Work Distribution

Student	Individual App	Parts Done
Ahmed Bahaa	To-Do List	1,2
Osama Ahmed	Smart Budget Manager	1,2
Mahmoud Tarek	Quiz App	1,3

Task 1

Team Learning Table

Team Member	Programming Language	Hours Spent Learning	Learning Resources
Osama Ahmed	Java	4hours	Learn Java From Scratch Java Documentation
Ahmed Bahaa	Java	5 hours	Learn Java From Scratch Java Documentation
Mahmoud Tarek	Java	3 hours	Learn Java From Scratch Java Documentation

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Evidence of Implementation

Team Member	Individual App	Screenshots (6–10)	Video Link
Osama Ahmed	Smart Budget Manager	Task1 - Java scr…	Smart Budget M…
Ahmed Bahaa	To-Do List	Task 1 Images 2…	To-Do List By J…
Mahmoud Tarek	Quiz App	Images-Ass-1-S…	Quiz App - Soft…

The code

1. Smart Budget Manager Programme code

```
iist<Transaction> transactions = new ArrayList<>();
    static Scanner input = new Scanner(System.in);

    public static void main(String[] args) {

        System.out.println("================================");
        System.out.println("      SMART BUDGET MANAGER");
        System.out.println("================================");

        menu();
        input.close();
    }

    // ================= MENU =================
    static void menu() {
        int choice;

        do {
            System.out.println("\n1- Add Income 💰");
            System.out.println("2- Add Expense 🧾");
            System.out.println("3- Show Report 📋");
            System.out.println("4- Search Category 🔎");
            System.out.println("5- Exit 🔚");
            System.out.print("Enter your choice: ");

            choice = input.nextInt();

            switch (choice) {
                case 1:
                    addIncome();
                    break;
                case 2:
```

```
                    addExpense();
                    break;
                case 3:
                    showReport();
                    break;
                case 4:
                    searchCategory();
                    break;
                case 5:
                    System.out.println("Exiting... 👋");
                    break;
                default:
                    System.out.println("Invalid choice ❌");
            }

        } while (choice != 5);
    }

    // ================= ADD INCOME =================
    static void addIncome() {
        System.out.print("Enter Income Amount: ");
        int amount = input.nextInt();

        while (amount <= 0) {
            System.out.print("Enter valid amount ❌: ");
            amount = input.nextInt();
        }

        System.out.print("Enter Category: ");
        String category = input.next();

        Transaction t = new Transaction(amount, "Income",
category);
        transactions.add(t);

        totalIncome += amount;
```

```
        System.out.println("✅ Income added successfully!");
    }

    // ================= ADD EXPENSE =================
    static void addExpense() {
        System.out.print("Enter Expense Amount: ");
        int amount = input.nextInt();

        while (amount <= 0) {
            System.out.print("Enter valid amount ❌: ");
            amount = input.nextInt();
        }

        System.out.print("Enter Category: ");
        String category = input.next();

        Transaction t = new Transaction(amount, "Expense",
category);
        transactions.add(t);

        totalExpense += amount;

        System.out.println("✅ Expense added successfully!");
    }

    // ================= SHOW REPORT =================
    static void showReport() {

        double netBalance = totalIncome - totalExpense;

        System.out.println("\n====== BUDGET REPORT ======");
        System.out.println("Total Income  : " + totalIncome);
        System.out.println("Total Expense : " + totalExpense);
        System.out.println("Net Balance   : " + netBalance);

        System.out.println("\n--- All Transactions ---");
        for (Transaction t : transactions) {
```

```
            System.out.println(t.type + " - " + t.amount + " - "
+ t.category);
        }

        // Largest Expense
        int maxExpense = 0;
        String maxCategory = "";

        for (Transaction t : transactions) {
            if (t.type.equals("Expense") && t.amount >
maxExpense) {
                maxExpense = t.amount;
                maxCategory = t.category;
            }
        }

        if (maxExpense > 0) {
            System.out.println("\nLargest Expense: " + maxExpense
+ " in " + maxCategory);
        } else {
            System.out.println("\nNo expenses recorded yet.");
        }
    }

    // ================= SEARCH =================
    static void searchCategory() {

        System.out.print("Enter Category to search 🔎: ");
        String searchCat = input.next();

        boolean found = false;

        System.out.println("\n--- Results ---");

        for (Transaction t : transactions) {
            if (t.category.equalsIgnoreCase(searchCat)) {
                System.out.println(t.type + " - " + t.amount);
```

```
                found = true;
            }
        }

        if (!found) {
            System.out.println("No transactions found ❌");
        }
    }
}
```

2. To-Do List Main.Java

```
import java.util.ArrayList;
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);
        ArrayList<Task> tasks = new ArrayList<>();

        int nextId = 1;
        int choice = 0;

        while (choice != 5) {

            System.out.println("\n1. Add Task");
            System.out.println("2. View Tasks");
            System.out.println("3. Mark Task as Done");
            System.out.println("4. Delete Task");
            System.out.println("5. Exit");

            System.out.print("Choose: ");
```

```
            choice = input.nextInt();
            input.nextLine();

            if (choice == 1) {

                System.out.print("Enter task title: ");
                String title = input.nextLine();

                System.out.print("Enter due date: ");
                String dueDate = input.nextLine();

                System.out.println("Select priority:");
                System.out.println("1. Low");
                System.out.println("2. Medium");
                System.out.println("3. High");

                System.out.print("Enter number (1-3): ");
                int priority = input.nextInt();
                input.nextLine();

                if (priority < 1 || priority > 3) {
                    System.out.println("Invalid priority. Setting
to Low.");
                    priority = 1;
                }

                Task t = new Task(nextId, title, dueDate,
priority, false);
                tasks.add(t);
                nextId++;

                System.out.println("Task added!");
            }

            else if (choice == 2) {

                if (tasks.size() == 0) {
```

```
                    System.out.println("No tasks available.");
                } else {

                    tasks.sort((t1, t2) -> t2.getPriorityValue()
- t1.getPriorityValue());

                    for (int i = 0; i < tasks.size(); i++) {
                        System.out.println(tasks.get(i));
                    }
                }
            }

            else if (choice == 3) {

                System.out.print("Enter task ID: ");
                int id = input.nextInt();
                input.nextLine();

                boolean found = false;

                for (Task t : tasks) {
                    if (t.id == id) {
                        t.isCompleted();
                        found = true;
                        System.out.println("Task marked as
done!");
                        break;
                    }
                }

                if (!found) {
                    System.out.println("Task not found.");
                }
            }

            else if (choice == 4) {
```

```
                System.out.print("Enter task ID to delete: ");
                int id = input.nextInt();
                input.nextLine();

                boolean removed = false;

                for (int i = 0; i < tasks.size(); i++) {
                    if (tasks.get(i).id == id) {
                        tasks.remove(i);
                        removed = true;
                        System.out.println("Task deleted!");
                        break;
                    }
                }
                if (!removed) {
                    System.out.println("Task not found.");
                }
            }
        }
        input.close();
        System.out.println("Goodbye!");
    }
}
```

3. Quiz App Code

```
package Ass;
import java.util.Scanner ;
class Question{
     Scanner Input = new Scanner(System.in) ;

     private String userName = "unKonwn";

     private int score = 0 ;

     private String[][] questions =
          {
                    {" (1) What is the size of int in Java?", "8
bits", "16 bits", "32 bits", "64 bits"} ,
                    {" (2) Which keyword creates an object?",
"class", "new", "void", "int"},
                    {" (3) What does OOP stand for?", "Open Oriented
Program", "Object Oriented Programming", "Old Old Program",
"None"} ,
                    {" (4) Which loop checks condition after
executing?", "for", "while", "do-while", "if"} ,
                    {" (5) What is the default value of int in
Java?", "null", "1", "0", "-1"}
          };

     private String[][] answers =
          {
                    {"Q1", "false", "false", "true", "false"},
                    {"Q2", "false", "true", "false", "false"},
                    {"Q3", "false", "true", "false", "false"},
                    {"Q4", "false", "false", "true", "false"},
                    {"Q5", "false", "false", "true", "false"},
          };

     public Question() {
          System.out.println("Welcome In Quiz App With Java
```

```
Programming");
        System.out.print("Enter Your Name : ");
        String name = Input.nextLine() ;
        this.userName = name ;
    }

    public void startQuestion() {
        System.out.println("Welcome " + this.userName +" We Have
" + this.questions.length + " Question In Our App ");

        for (int i = 0 ; i < this.questions.length ; i++) {

            System.out.println(this.questions[i][0]); // display
the question

            for (int j = 1 ; j <= this.answers.length -1; j ++)
{
                System.out.println("(" + j + ") "+
this.questions[i][j]); // display the options
            }

            System.out.print("Your Answer is : ");

            int ans = Input.nextInt() ;

            if (this.answers[i][ans].equals("true")) {  // check
if the answer is correct
                ++score;
            }

        }// end of main loop Block
    } // end of startQuestion Function Block

    public void showScore() {
        System.out.println("Your Score From " +
this.questions.length + " Questions is " + this.score);
    }
```

```
}// end of Question Class

public class QuizApp {
    public static void main(String[] args) {

        Question q = new Question() ;
        q.startQuestion();
        q.showScore();
    }

}
```

Repo Link: [CS251-Software-Engineering-Assignment1]

Task 2

What are LCNC Platforms?

Low-Code / No-Code (LCNC) platforms are tools that let people build apps and software with little or no coding. Instead of writing long lines of code, you use simple visual tools like drag-and-drop, ready-made templates, and built-in features. This makes it easier and faster to create websites, mobile apps, or business systems, even if you're not a professional developer. Companies use LCNC to save time, reduce cost, and allow non-technical staff to build solutions on their own. Some popular examples are OutSystems, Mendix, and Microsoft Power Apps.

Base 44

What It Can Do

Base44 is an AI app builder where you describe your idea in normal language, and the system creates the app for you. It builds the interface, database, backend logic, user login system, and even hosting. It's mainly focused on web apps. The goal is speed you can go from idea to working app very fast.

Benefits:

- Very easy for beginners.
- Fast setup, great for MVPs and testing ideas.
- Hosting and backend are included, so you don't need technical setup.
- Clean and simple interface.

Quality of the Systems It Produces

The review highlights how fast and simple it is, but speed sometimes comes at the cost of stability. For simple apps, it works well. For more complex systems, users may face bugs, limits in customization, or AI misunderstandings. It's strong for prototypes, but large business apps may require more control than it offers.

Potential

Base44 has strong potential because AI-driven building is growing fast. If stability and control improve, it could become very powerful for startups and solo founders.

Will It Replace Developers?

Not fully. It reduces the need for coding in simple projects, but complex apps still require deeper logic, optimization, security handling, and custom features that professional developers provide.

Bubble

What It Can Do

Bubble is a no-code platform where you visually build web apps using drag-and-drop tools. You design pages, create workflows, manage databases, and connect APIs. It supports complex logic and detailed customization. It's used for marketplaces, dashboards, SaaS apps, and more.

Benefits

- Very flexible and customizable.
- Strong plugin marketplace.
- Can scale to support real businesses.
- More control over data structure and workflows.

Quality of the Systems It Produces

Apps built with Bubble are usually more structured and scalable compared to many AI-only builders. It has a learning curve, but the final product can feel close to traditionally coded apps. Many startups run full businesses on Bubble.

Potential

Bubble has strong long-term potential because it balances ease of use with deep control. It continues to add AI features while keeping manual control available.

Will It Replace Developers?

For small and medium projects, it can reduce the need for hiring a developer. But for high-performance systems, large-scale platforms, or very custom solutions, experienced developers are still needed.

Bubble's Strengths

- Extremely flexible deeper control over data structures, user flows, and integrations.
- Large plugin ecosystem so you can extend what your app can do.
- Better support for team collaboration and scaling apps to real business use.

Comparison

Quality of the Systems They Produce

- **Base44** can generate a working app quickly, but the output sometimes needs polishing especially for design, performance, or complex features. Extra manual tweaks are often needed.
- **Bubble** produces systems that feel more professional and customizable. You'll still refine UI, flows, and data models, but it's closer to what businesses need for real users.

Real-world testers note that **Base44** might finish simple apps faster, while **Bubble's** tools let you build more polished and fully-featured products.

Can They Replace Developers?

Not fully. These tools are powerful, but they don't replace the depth of a human developer at least not yet.

Here's why:

What they *can* replace:

- Boilerplate setup and repetitive tasks.
- Basic backend systems like authentication and data storage.
- Fast prototyping for validating ideas.

What they *can't* replace:

- Deep architectural planning for large or complex systems.
- Writing custom algorithms, integrations, or performance-critical code.
- Handling edge-case logic and long-term maintainability.

Developers still play a role, especially when apps need unique features, custom security, or complex integrations. These tools shift developer effort from routine coding to higher-level design and problem solving instead of fully taking jobs away.

Feature / Area	Base44	Bubble
Type	AI-driven app builder	Visual no-code web app builder
Ease of Use	Very easy	Moderate (steeper learning curve)
Speed of Building	Very fast	Fast but more hands-on
Best For	Prototypes & simple apps	Full-featured web apps
Customization	Limited	High
Backend & Hosting	Built-in	Built-in
Support for Complex Logic	Basic to medium	Strong
Plugin / Integrations	Limited	Large marketplace
Quality of Output	Good for simple work	Stronger and more scalable
Developer Replacement Potential	Reduces beginner coding	Reduces coding for many cases, but not full replacement
Ideal Users	Beginners / small projects	Founders and teams building real products

Screenshots Link: Task 2 Images
Video Link: Task 2 Video.mp4

Task 3

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1 - Introduction :

In the last few years the way people use financial services has changed a lot. Before people had to go to the bank to send money pay bills or open an account. Now most of these things can be done from a mobile phone in a few minutes. This change happened because of FinTech.

FinTech means Financial Technology. It is the use of technology to provide financial services in an easier faster and cheaper way. FinTech helps people pay online transfer money get loans invest and buy products in installments using mobile applications or websites. Today FinTech is very important around the world and Egypt is becoming one of the growing countries in this field.

---

2 - Overview of the Global FinTech Industry :

Around the world FinTech has changed the financial system. Many companies now provide digital services instead of traditional banks. For example companies like PayPal Stripe and Revolut offer online payments digital banking and money transfer services in a simple and fast way.

FinTech includes different areas such as digital payments online lending buy now pay later services digital banks insurance technology and blockchain. The main goal of FinTech is to make financial services easier and available for more people especially those who do not use banks.

Because of technology people can now control their money using their phones without visiting a bank branch.

---

3 - FinTech in Egypt Market Overview :

In Egypt the FinTech sector has grown very fast in the last five years. According to the Egyptian FinTech Landscape 2024 report there are more than 177 FinTech startups and payment service providers in Egypt. This number shows strong growth compared to the past.

The sector increased more than five times in five years which shows that investors and the government are interested in this field. FinTech also represents a large percentage of startup investments in Egypt.

The Central Bank of Egypt supports digital transformation and financial innovation. This support helped many startups grow and expand in the Egyptian market.

---

4 - Economic and Investment Environment :

The Egyptian economy faced many challenges in recent years because of global events like the pandemic and international conflicts. These events affected the currency and investment flow.

In 2024 Egypt changed its exchange rate system to improve economic stability and attract more foreign investment. This step helped create a better environment for startups including FinTech companies.

However many startups still face problems in getting funding in the early stage. After passing the early stage it becomes easier to get bigger investments.

---

5 - Geographic Distribution and Infrastructure

Most FinTech companies in Egypt are located in Cairo and Giza because these areas have strong economic activity and better internet services. Alexandria also has some companies but the percentage is smaller.

Upper Egypt and Sinai have weaker internet infrastructure and this makes it harder for FinTech services to spread in these regions.

If the government improves digital infrastructure in rural areas more people will be able to use digital financial services and this will increase financial inclusion in Egypt.

---

6 - Leading Egyptian FinTech Companies :

FinTech in Egypt grew very fast in the last few years. Many companies helped change the way people use money and financial services. These companies made payments easier and reduced the need to use cash.

Fawry : is one of the biggest payment companies in Egypt. People use it to pay electricity bills water bills mobile recharge and many other services. It has payment machines and stores all over the country. Because of Fawry many people can pay easily without going to the bank.
ValU : allows people to buy products and pay in installments. This is called buy now pay later. Customers can buy phones home appliances and other products and pay over time. It made shopping easier for many people.
Khazna : helps people who do not have bank accounts. It gives small loans and allows bill payments through a mobile application. It supports low income workers and helps them manage their money better.
Money Fellows : uses technology to organize traditional saving groups. Instead of saving money in an informal way people can now join digital saving groups safely through the app.
Paymob : helps businesses accept online payments. Stores can accept card payments and mobile wallet payments easily. This supports online shopping in Egypt.

These companies helped Egypt move toward a digital and cashless economy.

---

7 - Ecosystem Support in Egypt :

FinTech companies in Egypt are supported by important organizations.

The Central Bank of Egypt supports digital payments and financial technology. It creates rules to organize the sector.

The Financial Regulatory Authority controls non banking financial services like insurance and microfinance.

There are also programs that help startups grow such as :FinTech Egypt , Flat6Labs ,AUC Venture Lab, Falak Startups

These programs give training mentorship and sometimes funding to new companies.

Investors also started to put more money in Egyptian FinTech companies because they believe the market will continue to grow.

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8 - Halal and Non Halal FinTech Activities :

In Islamic finance some financial activities are allowed and some are not allowed.

Halal Activities

Halal activities include

Islamic financing like Murabaha and Musharakah
Profit sharing investments
Islamic insurance known as Takaful
Financing based on real assets not interest

These models avoid interest and unfair practices.

Non Halal Activities

Non halal activities include

Loans with interest
Credit systems that depend on compound interest
Very risky speculative trading

Many people in Egypt prefer Shariah compliant financial services so Islamic FinTech has good potential in the country.

---

9 - Challenges Facing FinTech in Egypt :

Even though the sector is growing it still has some problems.

Regulations can sometimes be complicated.

Some people still prefer cash and do not trust digital payments.

Cybersecurity is important because companies must protect customer data.

Some startups find it hard to get large investments.

Internet service in rural areas is not as strong as in big cities.

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10 - Conclusion :

FinTech in Egypt is growing very fast. It helps people pay shop and manage money in an easier way.

Companies like Fawry ValU and Paymob changed financial services in Egypt.

There are still some challenges but the future looks positive. Egypt has a strong chance to become an important FinTech hub in the Middle East and North Africa region.

References

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